

Analytic Task 2 – Discipline / Cards

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Research Question Is there a statistically significant difference in the average number of cards (yellow + red) per match between the Group Stage and the Knockout Stage (Round of 32 onward) of the 2026 FIFA World Cup?

Data Source Card data was collected from FBref, from the Miscellaneous Stats section of each match report (CrdY / CrdR per team):

FBref – FIFA World Cup 2026 Stats: <https://fbref.com/en/comps/1/World-Cup-Stats>

FBref doesn't provide a bulk export for per-match stats, so each match had to be looked up individually and the numbers recorded by hand. The collected data is in raw_data.csv.

The underlying match statistics displayed on FBref are provided by Stats Perform (SR). Per FBref's data usage terms, this is acknowledged here: data provided by Stats Perform (<https://www.statsperform.com/>).

Data Preparation: The population is all 104 matches of the 2026 World Cup: 72 Group stage matches and 32 Knockout matches (Round of 32, Round of 16, Quarterfinals, Semi-finals, Third-place match, Final).

For each match I calculated:

Total Cards = Yellow Cards (Home + Away) + Red Cards (Home + Away)

I also added a second variable, Cards per 90, to adjust for match length:

Cards per 90 = (Total Cards / Minutes Played) × 90

This matters because two matches in the sample (Belgium v Senegal, Australia v Egypt) went to extra time and ran 120 minutes instead of 90, so they naturally had more time for cards to be shown. The data was checked for missing values before analysis.

Sampling I used a stratified random sample of 35 matches from the population of 104, stratified by stage so the smaller Knockout group would still be well represented. This gave 20 Group stage matches and 15 Knockout matches. Random seed 42 was used so the sample can be reproduced. The sample is in raw_data.csv.

Descriptive Statistics

Statistic	Group Stage	Knockout Stage	Sample size (n)
Mean	3.150	2.733	20 15
Std deviation	1.927	2.052	
Minimum	0	0	
Maximum	7	8	

Group stage matches averaged slightly more cards than Knockout matches. Both stages had at least one match with zero cards.

Since both sample sizes are under 30, I checked the normality assumption using a Shapiro-Wilk test rather than just assuming it: Group $p = 0.169$, Knockout $p = 0.174$. Both are above 0.05, so there's no strong evidence against normality.

Confidence Interval 95% confidence intervals for mean cards per match:

Group stage: [2.248, 4.052] Knockout stage: [1.597, 3.870]

The two intervals overlap a lot, which already suggests the difference might not hold up statistically.

Hypothesis Test An independent two-sample t-test was used to compare the two stages.

H_0 : mean cards (Group) = mean cards (Knockout) H_1 : mean cards (Group) $\neq$ mean cards (Knockout)

I used Welch's t-test since the two sample sizes aren't equal (20 vs 15):

$t = 0.6102$, $df = 29.23$, $p = 0.5465$

Since $p > 0.05$, I failed to reject H_0 . For comparison, I also calculated df the simpler way taught in class ($df = \min(n_1-1, n_2-1) = 14$), which gives $p = 0.5515$ same conclusion either way.

As a check, I reran the test using Cards per 90 instead of the raw total, to see if adjusting for the two extra-time matches changed anything: $t = 0.7284$, $df = 29.07$, $p = 0.4722$. Still the same conclusion.

Conclusion: There's no statistically significant difference in average cards per match between the Group stage and the Knockout stage. The mean difference was 0.417 cards per match (slightly higher in Group stage), but the 95% confidence interval for that difference, [-0.98, 1.81], includes zero, so it's not a reliable difference.

This held up even after adjusting for match length, so the result isn't just an artefact of the two extra-time matches. Since the sample was drawn using proper stratified random sampling rather than picking convenient matches, this should generalise reasonably well to the tournament. One limitation: this is based on 35 out of 104 matches, and card counts are only one part of "discipline" fouls and VAR decisions weren't included.

Files:

1. discipline_analysis.py – Python code for data wrangling, descriptive statistics, the confidence interval, and the t-test.
2. raw_data.csv – Sample of 35 matches with card counts and match length.

3. processed_data.csv – Cleaned dataset with Total Cards and Cards per 90 added, produced by the script cards.
4. boxplot.png – Boxplot comparing total cards per match, Group vs Knockout.